Applications of Fuzzy Sets in Industrial Engineering: A Topical Classification

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Summary: A rational approach toward decision-making should take into account human subjectivity, rather than employing only objective probability measures. This attitude towards the uncertainty of human behavior led to the study of a relatively new decision analysis field: Fuzzy decision-making. Fuzzy systems are suitable for uncertain or approximate reasoning, especially for the system with a mathematical model that is difficult to derive. Fuzzy logic allows decision-making with estimated values under incomplete or uncertain information. A major contribution of fuzzy set theory is its capability of representing vague data. Fuzzy set theory has been used to model systems that are hard to define precisely. As a methodology, fuzzy set theory incorporates imprecision and subjectivity into the model formulation and solution process. Fuzzy set theory represents an attractive tool to aid research in industrial engineering (IE) when the dynamics of the decision environment limit the specification of model objectives, constraints and the precise measurement of model parameters. This chapter provides a survey of the applications of fuzzy set theory in IE.

Key words: Industrial engineering, fuzzy sets, research areas, decision-making, multiple criteria, quality management, production planning, manufacturing, ergonomics, artificial intelligence, engineering economics.

1 Introduction

Fuzzy set theory, which was founded by Zadeh (1965), has emerged as a powerful way of representing quantitatively and manipulating the imprecision in decision-making problems. Fuzzy sets or fuzzy numbers can appropriately represent imprecise parameters, and can be manipulated through different operations on fuzzy sets or fuzzy numbers. Since imprecise parameters are treated as imprecise values instead of precise ones, the process will be more powerful and its results more credible. Fuzzy set theory has been studied extensively over the past 40 years. Fuzzy set theory is now applied to problems in engineering, business, medical and related

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health sciences, and the natural sciences. In an effort to gain a better understanding of the use of fuzzy set theory in industrial engineering (IE) and to provide a basis for future research, a literature review of fuzzy set theory in IE has been conducted. Over the years there have been successful applications and implementations of fuzzy set theory in IE. Fuzzy set theory is being recognized as an important problem modeling and solution technique. The use of fuzzy set theory as a methodology for modeling and analyzing decision systems is of particular interest to researchers in IE due to fuzzy set theory's ability to quantitatively and qualitatively model problems which involve vagueness and imprecision. Fuzzy set theory can be used to bridge modeling gaps in descriptive and prescriptive decision models in IE.

There are many national and international associations and societies to support the development of the fuzzy set theory. The *International Fuzzy Systems Association* (IFSA) is a worldwide organization dedicated to the support, development and promotion of the fundamental issues of fuzzy theory related to (a) sets, (b) logics, (c) relations, (d) natural languages, (e) concept formation, (f) linguistic modeling, (g) vagueness, (h) information granularity, etc. and their applications to (1) systems modeling, (2) system analysis, (3) diagnosis, (4) prediction and (5) control in decision support systems in management and administration of human organizations as well as electro-mechanical systems in manufacturing and process industries. IFSA organizes and encourages participation in open discussion session on the future directions and restructuring of fuzzy theory in IFSA congresses. Furthermore, IFSA publishes the *International Journal of Fuzzy Sets and Systems* and sponsors workshops and conferences on fuzzy theory. Japan Society for Fuzzy Theory and Systems (SOFT) was established in 1989. SOFT have 1,670 individual members and 74 company members, publishes an official bimonthly journal and organizes fuzzy systems symposiums. There are 8 regional branches and 8 research groups in SOFT. The *North American Fuzzy Information Processing Society* (NAFIPS) was established in 1981 as the premier fuzzy society in North America. Its purpose is to help guide and encourage the development of fuzzy sets and related technologies for the benefit of mankind. *Berkeley Initiative in Soft Computing (BISC) Program* is the world-leading center for basic and applied research in soft computing. The principal constituents of soft computing are fuzzy logic, neural network theory and probabilistic reasoning, with the latter subsuming belief networks, evolutionary computing including DNA computing, chaos theory and parts of learning theory. Some of the most striking achievements of BISC Program are: fuzzy reasoning (set and logic), new soft computing algorithms making intelligent, semi-unsupervised use of large quantities of complex data, uncertainty analysis, perception-based decision analysis and decision support systems for risk analysis and management, computing with words, computational theory of perception, and precisiated natural language. *Spanish Association of Fuzzy Logic and Technologies* promotes and disseminates the methods, techniques and developments of Fuzzy Logic and Technologies; Establish relations with other national or international Associations with similar aims; Organizes seminars and round tables on Fuzzy Logic and Technologies. The European Society for Fuzzy Logic and Technology (EUSFLAT) was established in 1998. The main goal of EUSFLAT is to represent the European fuzzy community of IFSA. Hungarian Fuzzy Society was established in 1998. EUROFUSE Working Group on Fuzzy Sets of EURO was established in 1975. The purpose of EUROFUSE is to communicate and promote the knowledge of the theory of fuzzy sets and related areas and their applications. In 1985, the Group constituted itself as the European Chapter of IFSA. *EUropean Network on Intelligent TEchnologies for Smart Adaptive Systems* (EUNITE) has started on 1 January 2001. It is funded by the *Information Society Technologies Programme* (IST) within the European Union's Fifth RTD Framework Programme.

Some bibliographies related to the interest areas of IE are given as follows. Gaines and Kohout (1977), Kandel and Yager (1979), Kandel (1986), and Kaufmann and Gupta (1988) show the applications of fuzzy set theory in general. The bibliographies by Zimmerman (1983), Chen and Hwang (1992) and Lai and Hwang (1994) review the literature on fuzzy sets in operations research, fuzzy multiple attribute decision making (Fuzzy MADM) and fuzzy multiple objective decision-making (Fuzzy MODM) respectively. Maiers and Sherif (1985) review the literature on fuzzy industrial controllers and provide the applications of fuzzy set theory to many subject areas including decision making, economics, engineering and operations research. Karwowski and Evans (1986) identify the potential applications of fuzzy set theory to the following areas of IE: new product development, facilities location and layout, production scheduling and control, inventory management, quality and cost benefit analysis. In their book, Evans et al. (1989) combined contributions from world renowned experts on the various topics covered such as: traditional IE techniques; ergonomics; safety engineering; human performance measurement; manmachine systems and fuzzy methodologies. Klir and Yuan (1995) and Zimmermann (1996) reviewed the applications of fuzzy set theory and fuzzy logic.

In this paper, we review the literature and consolidate the main results on the application of fuzzy set theory to IE. The purpose of this paper is to: (i) review the literature; (ii) classify the literature based on the application of fuzzy set theory to IE; and, (iii) identify future research directions. This paper is organized as follows. Section 2 introduces a classification scheme for fuzzy research in IE. Section 3 reviews previous research on fuzzy set theory and IE. The conclusions to this study are given in Section 4.

2 Classification Scheme for Fuzzy Set Theory Applications in IE Research

Table 1 illustrates a classification scheme for the literature on the application of fuzzy set theory in IE. In this table, many categories are defined and the frequency of citations in each category is identified. Artificial Intelligence resulted in the largest number of citations (50), followed by Multiple Criteria Decision-making (42), and statistical decision-making (32) and manufacturing (22). This survey is restricted to research on the application of fuzzy sets to IE decision problems. In our study, a total of 473 citations on the application of fuzzy set theory in IE were found (see Table 1 and Table 2). All of these studies come from the departments of IE of the universities. Eight journals, *Fuzzy Sets and Systems, Information Sciences, Computers & Industrial Engineering, Computers & Mathematics with Applications*, *European Journal of Operational Research*, *International Journal of Production Economics*, and *International Journal of Approximate Reasoning* accounted for 70 percent of the citations. Table 3 gives the main books published on fuzzy applications of IE topics.

Table 1. Classification Scheme for Fuzzy Set Research in IE (1972 – 2005)

Research topic	Number of citations
Artificial Intelligence	50
Multiple Criteria Decision-making	42
Statistical decision-making	32
Manufacturing	22
Classification / Clustering	19
Optimization	19
General Modeling	18
Expert systems	15
Decision-making	14
Control	13
Ergonomics	12
Linear programming	12
Mathematical Programming	10
Engineering economics	9
Decision support system	8
Group decision-making	8
Information Systems	7
Production management	6
Production Systems	6
Project management	5
Technology Management	4
Quality management	3

Forecasting	2
Simulation	2
Risk assessment	1
Others	134
Total	473

Table 2. Summary of Journal Citations on Fuzzy Set Theory in IE (1972 – 2005)

Journal Title	Frequency
Accident Analysis & Prevention	1
Advances in Engineering Software and Power Systems: A Fuzzy Network Model	1
Applied Mathematics and Computation	9
Applied Mathematics Letters	2
Applied Soft Computing	4
Applied Thermal Engineering	1
Artificial Intelligence In Medicine	2
Automatica	2
Computers & Electrical Engineering	1
Computers & Industrial Engineering	33
Computers & Mathematics with Applications	35
Computers & Operations Research	8
Computers in Industry	7
Conservation and Recycling	1
Control Engineering Practice	3
Decision Support Systems	3
Displays	1
Ecological Modelling	1
Energy Conversion and Management	1
Engineering Applications of Artificial Intelligence	7
Ergonomics	3
European Journal of Operational Research	37
Expert Systems with Applications	9

Journal Title	Frequency
Fuzzy Group Decision-Making for Facility Location Selection	1

Fuzzy prediction of maize breakage	1
Fuzzy Sets and Systems	177
Information and Control	1
Information Sciences	22
International Journal of Approximate Reasoning	11
International Journal of Human-Computer Studies	2
International Journal of Industrial Ergonomics	5
International Journal of Machine Tools and Manufacture	2
International Journal of Production Economics	14
International Journal of Project Management	2
International Transactions in Operational Research	1
Journal of Agricultural Engineering Research	1
Journal of Computational and Applied Mathematics	1
Journal of Environmental Management	1
Journal of Human Ergology	1
Journal of Materials Processing Technology	3
Journal of Mathematical Analysis and Applications	8
Mathematical and Computer Modelling	3
Mathematical Modelling	1
Mathematical Models Used for Adaptive Control of Machine Tools	1
Mathematics and Computers in Simulation	1
Mechatronics	3
Methods of Information In Medicine	1
Microelectronics and Reliability	2
Neural Networks	3
Neural Networks: The Official Journal of the International Neural Network Society	2
Neurocomputing	1
Nuclear Engineering and Design	1
Omega	3
Pattern Recognition	4
Pattern Recognition Letters	4
Reduction and Axiomization of Covering Generalized Rough Sets	1
Reliability Engineering & System Safety	1
Renewable Energy	1
Journal Title	**Frequency**

Robot Selection Decision Support System: A Fuzzy Set Approach	1
Robotics and Autonomous Systems	6
Robotics and Computer-Integrated Manufacturing	4
Technological Forecasting and Social Change	1
Trends in Pharmacological Sciences	1
Total	472

Table 3. Some books published on fuzzy IE

Authors	Contents
Bector and Chandra (2005)	This book presents a systematic and focused study of the application of fuzzy sets to two basic areas of decision theory, namely Mathematical Programming and Matrix Game Theory. Most of the theoretical results and associated algorithms are illustrated through small numerical examples from actual applications.
Dompere (2004)	This monograph is devoted to the development of value theory of computable general prices in costbenefit analysis under fuzzy rationality. The book demonstrates the use of fuzzy decision algorithms and logic to develop a comprehensive and multidisciplinary cost-benefit analysis by taking advantage of current scientific gains in fuzziness and soft computing.
Aliev et al. (2004)	This monograph provides a self-contained exposition of the foundations of soft computing, and presents a vast compendium of its applications to business, finance, decision analysis and economics. The applications range from transportation and health case systems to intelligent stock market prediction, risk management systems, and e-commerce.
Onwubolu and Babu (2004)	The book describes a variety of the novel optimization techniques which in most cases outperform the standard optimization techniques in many application areas. New Optimization Techniques in Engineering reports applications and results of the novel optimization techniques considering a multitude of practical problems in the different engineering disciplines.

Authors	Contents
Niskanen (2004)	This book includes an in-depth look at soft computing methods and their applications in the human sciences, such as the social and the behavioral sciences. The powerful application areas of these methods in the human sciences are demonstrated.
Verdegay (2003)	The aim of this book is to show how Fuzzy Sets and Systems can help to provide robust and adaptive heuristic optimization algorithms in a variety of situations. The book presents the state of the art and gives a broad overview on the real practical applications that Fuzzy Sets have, based on heuristic algorithms.
Yu and Kacprzyk (2003)	This book provides a comprehensive coverage of up-to-date conceptual frameworks in broadly perceived decision support systems and successful applications. Different from other existing books, this book predominately focuses on applied decision support with soft computing. Areas covered include planning, management finance and administration in both the private and public sectors.
Castillo and Melin (2003)	The book describes the application of soft computing techniques and fractal theory to intelligent manufacturing. The text covers the basics of fuzzy logic, neural networks, genetic algorithms, simulated annealing, chaos and fractal theory. It also describes in detail different hybrid architectures for developing intelligent manufacturing systems for applications in automated quality control, process monitoring and diagnostics, adaptive control of nonlinear plants, and time series prediction.
Buckley et al. (2002)	The book aims at surveying results in the application of fuzzy sets and fuzzy logic to economics and engineering. New results include fuzzy non-linear regression, fully fuzzified linear programming, fuzzy multi-period control, fuzzy network analysis, each using an evolutionary algorithm; fuzzy queuing decision analysis using possibility theory; fuzzy hierarchical analysis using an evolutionary

	algorithm. Other important topics covered are fuzzy inputoutput analysis; fuzzy mathematics of finance; fuzzy PERT (project evaluation and review technique).

Authors	Contents
Carlsson and Fuller (2002)	This book starts with the basic concepts of fuzzy arithmetics and progresses through fuzzy reasoning approaches to fuzzy optimization. Four applications of (interdependent) fuzzy optimization and fuzzy reasoning to strategic planning, project management with real options, strategic management and supply chain management are presented.
Yoshida (2001)	The book focuses on the recent dynamical development in fuzzy decision making. First, fuzzy dynamic programming is reviewed from a viewpoint of its origin and we consider its development in theory and applications. Next, the structure of dynamics in systems is considered in relation to fuzzy mathematical programming. Furthermore, some topics potentially related to dynamics are presented: financial management, fuzzy differential for fuzzy optimization of continuous-time fuzzy systems, and fuzzy ordering in multi-objective fuzzy systems.
Leiviskä (2001)	Applications of Soft Computing have recently increased and methodological development has been strong. The book is a collection of new interesting industrial applications introduced by several research groups and industrial partners. It describes the principles and results of industrial applications of Soft Computing methods and introduces new possibilities to gain technical and economic benefits by using this methodology.
Fodor et al. (2000)	In this book, leading scientists in the field address various theoretical and practical aspects related to the handling of the incompleteness. The problems discussed are taken from multi-objective linear programming, rationality considerations in preference modeling, non-probabilistic utility theory, data fusion, group decision making and

Authors	Contents
	multicriteria decision aid.
Yen et al. (1995)	It brings a functional understanding of the industrial applications of fuzzy logic in one self-contained volume. Application areas covered range from sensors, motors, robotics, and autonomous vehicles to process control and consumer products.
Klir and Yuan (1995)	This book both details the theoretical advances and considers a broad variety of applications of fuzzy sets and fuzzy logic. Part I is devoted to basics of fuzzy sets. Part II is devoted to applications of fuzzy set theory and fuzzy logic, including: the use of fuzzy logic for approximate reasoning in expert systems; fuzzy systems and controllers; fuzzy databases; fuzzy decision making; and engineering applications.
Evans et al. (1989)	Traditional IE techniques; ergonomics; safety engineering; human performance measurement; man-machine systems and fuzzy methodologies; locational analysis; job shop scheduling; financial planning; project management; data analysis; and fuzzy control of industrial processes; human factors modeling and organizational design, and, fuzzy set applications in operations research and decision analysis.

3 Fuzzy Set Theory and Industrial Engineering

Extensive work has been done on applying fuzzy set theory to research problems in IE. Using the classification scheme developed in Section 2, recent research findings in each area of IE will be reviewed.

3.1 Statistical Decision-Making

Statistics is a science of making decisions with respect to the characteristics of a group of persons or objects on the basis of numerical information obtained from a randomly selected sample of the group.

Provided with plenty of data (experience), data mining techniques are widely used to extract suitable management skills from the data. Nevertheless, in the early stages of a manufacturing system, only rare data can be obtained, and built scheduling knowledge is usually fragile. Using small data sets, Li *et al.* (2006) improve the accuracy of machine learning for flexible manufacturing system scheduling. They develop a data trend estimation technique and combine it with mega-fuzzification and adaptivenetwork-based fuzzy inference systems (ANFIS).

Clustering multimodal datasets can be problematic when a conventional algorithm such as k-means is applied due to its implicit assumption of Gaussian distribution of the dataset. Cho *et al.* (2006) propose a tandem clustering process for multimodal data sets. The proposed method first divides the multimodal dataset into many small pre-clusters by applying kmeans or fuzzy k-means algorithm. These pre-clusters are then clustered again by agglomerative hierarchical clustering method using KullbackLeibler divergence as an initial measure of dissimilarity.

Modarres et al. (2005) develop a mathematical programming model to estimate the parameters of fuzzy linear regression models with fuzzy output and crisp inputs. The method is constructed on the basis of minimizing the square of the total difference between observed and estimated spread values or in other words minimizing the least square errors.

Fuzzy relation is a crucial connector in presenting fuzzy time series model. However, how to obtain a fuzzy relation matrix to represent a timeinvariant relation is still a question. Based on the concept of fuzziness in Information Theory, Tsaur et al. (2005) apply the concept of entropy to measure the degrees of fuzziness when a time-invariant relation matrix is derived.

Wang and Kuo (2004) propose a method of factor analysis for a huge database so that not only the independence among the factors can be ensured and the levels of their importance can be measured, but also new factors can be discovered. To measure the independence of the factors, statistical relation analysis and the concept of fuzzy set theory are employed. A fuzzy set of 'the factors are almost dependent' is used to measure the degree of dependence between factors, and then through a hierarchical clustering procedure, the dependent factors are detected and removed. To measure the weights of importance for the independent factors, a supervised feedforward neural network is developed. In addition, they design a hierarchical structure to facilitate the extraction of new factors when the information of the system is not complete.

3.2 Manufacturing

The area of manufacturing systems is one of the main themes in industrial engineering. Many interest areas of industrial engineering such as job shops, machine efficiency, line balancing, job shop design, etc. are researched using fuzzy sets under vagueness.

Hop (2006) addresses the mixed-model line balancing problem with fuzzy processing time. A fuzzy binary linear programming model is formulated for the problem. This fuzzy model is then transformed to a mixed zero-one program. Due to the complexity nature in handling fuzzy computation, new approximated fuzzy arithmetic operation is presented. A fuzzy heuristic is developed to solve this problem based on the aggregating fuzzy numbers and combined precedence constraints. The general idea of this approach is to arrange the jobs in a sequence by a varying-section exchange procedure. Then jobs are allocated into workstations based on these aggregated fuzzy times with the considerations of technological constraint and cycle time limit.

Gholamian and Ghomi (2005) try to stimulate empirical research into the overall impacts of intelligent systems in manufacturing aspects. To reach this goal, a schema of intelligent applications is provided for each aspect as frame base structure, meaning the knowledge of intelligent applications in that specific aspect. Then, a semantic network is developed for intelligent manufacturing based on hierarchical structure of manufacturing systems to provide Meta knowledge of intelligent manufacturing applications. The analysis of semantic network indicates an increasing growth in application of soft computing in manufacturing aspects. Specially, fuzzy logic and its derivates have found more applications in recent years.

Chang and Liao (2005) present a novel approach by combining selforganizing map and fuzzy rule base for flow time prediction in semiconductor manufacturing factory. Flow time of a new order is highly related to the shop floor status; however, the semiconductor manufacturing processes are highly complicated and involve more than hundred of production steps. There is no governing function identified so far among the flow time of a new order and these shop flow status. Therefore, a simulation model which mimics the production process of a real wafer fab located in Hsin-Chu Science-based Park of Taiwan is built and flow time and related shop floor status are collected and fed into the self-organizing map for classification. Then, corresponding fuzzy rule base is selected and applied for flow time prediction. Genetic process is further applied to fine-tune the composition of the rule base.

Kahraman et al. (2004) propose some fuzzy models based on fuzzy present worth to measure manufacturing flexibility. The fuzzy models based on present worth are basically engineering economics decision models in which the uncertain cash flows and discount rates are specified as triangular fuzzy numbers. To build such a model, fuzzy present worth formulas of the manufacturing flexibility elements are formed. Flexibility for continuous improvement, flexibility for trouble control, flexibility for work force control, and flexibility for work-in-process control are quantified by using fuzzy present worth analysis. Formulas for both inflationary and noninflationary conditions are derived. Using these formulas, more reliable results can be obtained especially for such a concept like flexibility that is described in many intangible dimensions. These models allow experts' linguistic predicates about computer integrated manufacturing systems.

Chen et al. (2003) introduce a new tool path generation method to automatically subdivide a complex sculptured surface into a number of easy-to-machine surface patches; identify the favorable machining set-up/orientation for each patch; and generate effective 3-axis CNC tool paths for each patch.

3.3 Single Criterion and Multiple Criteria Decision-Making

Single criterion decision-making takes only one criterion into account such as profitability. Multiple criteria decision-making is divided into two components: MADM and MODM. MADM problems are widespread in real life decision situations. A MADM problem is to find a best compromise solution from all feasible alternatives assessed on multiple attributes, both quantitative and qualitative. In case of MODM, you have more than one objective function and some constraints. In the following, some recent works on this topic are summarized.

The advent of multiagent systems, a branch of distributed artificial intelligence, introduced a new approach to problem solving through agents interacting in the problem solving process. Chen and Wang (2005) address a collaborative framework of a distributed agent-based intelligence system to control and resolve dynamic scheduling problem of distributed projects for practical purposes. If any delay event occurs, the self-interested activity agent, the major agent for the problem solving of dynamic scheduling in the framework, can automatically cooperate with other agents in real time to solve the problem through a two-stage decision-making process: the fuzzy decision-making process and the compensatory negotiation process. The first stage determines which behavior strategy will be taken by agents while delay event occurs, and prepares to next negotiation process; then the compensatory negotiations among agents are opened related with determination of compensations for respective decisions and strategies, to solve dynamic scheduling problem in the second stage.

A MADM problem deals with candidate priority alternatives with respect to various attributes. MADM techniques are popularly used in diverse fields such as engineering, economics, management science, transportation planning, etc. Several approaches have been developed for assessing the weights of MADM problems, e.g., the eigenvector method, ELECTRE, TOPSIS, etc. Chen and Larbani (2006) obtained weights of a MADM problem with a fuzzy decision matrix by formulating it as a twoperson zero-sum game with an uncertain payoff matrix. Moreover, the equilibrium solution and the resolution method for the MADM game is also developed; these results are validated by a product development example of nano-materials.

The information axiom proposes the selection of the proper alternative that has minimum information. Selection of the convenient equipment under determined criteria (such as costs and technical characteristics) using the information axiom is realized by Kulak et al. (2005). The unweighted and weighted multi-attribute axiomatic design approaches developed in this paper include both crisp and fuzzy criteria. The fuzzy set theory has the capability of capturing the uncertainty under the conditions of incomplete, non-obtainable and unquantifiable information. These approaches are applied to the selection among punching machines while investing in a manufacturing system.

Beneficiation of coal refers to the production of wash coal from raw coal with the help of some suitable beneficiation technologies. The processed coal is used by the different steel plants to serve their purpose during the manufacturing process of steel. Chakraborty and Chandra (2005) deal with the optimal planning for blending raw coal of different grades used for beneficiation with a view to satisfy the requirements of the end users with desired specifications. The input specifications of coal samples are known whereas the output specifications are imprecise in nature. The aim of the work is to fix the level of the raw coal from different coal seams to be fed for beneficiation to meet the desired target of yield and ash percentage to maximum extent. Further, it is also desired by the decision-maker to restrict the input cost of raw coal to be fed for beneficiation. The problem is modeled as multicriteria decision-making problem with imprecise specifications. Fuzzy set theoretic approach is used and a corresponding model is developed.

MADM involves tradeoffs among alternative performances over multiple attributes. The accuracy of performance measures are usually assumed to be accurate. Most multiattribute models also assume given values for the relative importance of weights for attributes. However, there is usually some uncertainty involved in both of these model inputs. Outranking multiattribute methods have always provided fuzzy input for performance scores. Many analysts have also recognized that weight estimates also involve some imprecision, either through individual decision maker uncertainty, or through aggregation of diverging group member preferences. Many fuzzy multiattribute models have been proposed, but they have focused on identifying the expected value solution (or extreme solutions). Olson and Wu (2005) demonstrate how simulation can be used to reflect fuzzy inputs, which allows more complete probabilistic interpretation of model results.

Araz et al. (2005) propose a multi-objective covering-based emergency vehicle location model. The objectives considered in the model are maximization of the population covered by one vehicle, maximization of the population with backup coverage and increasing the service level by minimizing the total travel distance from locations at a distance bigger than a prespecified distance standard for all zones. Model applications with different solution approaches such as lexicographic linear programming and fuzzy goal programming are provided through numerical illustrations to demonstrate the applicability of the model.

Cochran and Chen (2005) develop a fuzzy set approach for multicriteria selection of object-oriented simulation software for analysis of production system. The approach uses fuzzy set theory and algebraic operations of fuzzy numbers to characterize simulation software so that the strength and weakness of each alternative can be compared. The linguistic input from experts and decision makers are fuzzified and the results of the fuzzy inference are translated back to linguistic explanation. Further, by aggregating decision maker's preference (weighting) to the expert's rating, a single number for each candidate software can be obtained. One can use this number as an index to choose the most suitable simulation software for the specific application.

3.4 Information Systems

Information system (IS) is a system for managing and processing information, usually computer-based. Also, it is a functional group within a business that manages the development and operations of the business's information. There are many works on IS using fuzzy set theory.

Wang et al. (2005) give an overview of applying fuzzy measures and relevant nonlinear integrals in data mining, discussed in five application areas: set function identification, nonlinear multiregression, nonlinear classification, networks, and fuzzy data analysis. In these areas, fuzzy measures allow us to describe interactions among feature attributes towards a certain target (objective attribute), while nonlinear integrals serve as aggregation tools to combine information from feature attributes. Values of fuzzy measures in these applications are unknown and are optimally determined via a soft computing technique based on given data.

Imprecise or vague importance as well as satisfaction levels of criteria may characterize the evaluation of alternative network security systems and may be treated by the fuzzy set theory. As fuzzy numbers are adopted, the fuzzy weighted averages (FWAs) may become a suitable computation for the fuzzy criteria ratings and fuzzy weightings. Chang and Hung (2005) propose to use the FWA approach that operates on the fuzzy numbers and obtains the fuzzy weighted averages during the evaluations of network security systems. The algorithm constructs a benchmark adjustment solution approach for FWA.

Maimon et al. (2001) present a novel, information-theoretic fuzzy approach to discovering unreliable data in a relational database. A multilevel information-theoretic connectionist network is constructed to evaluate activation functions of partially reliable database values. The degree of value reliability is defined as a fuzzy measure of difference between the maximum attribute activation and the actual value activation. Unreliable values can be removed from the database or corrected to the values predicted by the network. The method is applied to a real-world relational database which is extended to a fuzzy relational database by adding fuzzy attributes representing reliability degrees of crisp attributes. The highest connection weights in the network are translated into meaningful if, then rules. This work aims at improving reliability of data in a relational database by developing a framework for discovering, accessing and correcting lowly reliable data.

Demirli and Türkúen (2000) identify the robot's location in a global map from solely sonar based information. This is achieved by using fuzzy sets to model sonar data and by using fuzzy triangulation to identify robot's position and orientation. They obtain a fuzzy position region where each point in the region has a degree of certainty of being the actual position of the robot.

Tanaka et al. (1986) formulate a fuzzy linear programming problem and the value of information is discussed. If the fuzziness of coefficients is decreased by using information about the coefficients, they expect to obtain a more satisfactory solution than without information. With this view, the prior value of information is discussed. Using the sensitivity analysis technique, a method of allocating the given investigation cost to each fuzzy coefficient is proposed.

3.5 Expert Systems

Fuzzy expert systems have been successfully applied to a wide range of problems from different areas presenting uncertainty and vagueness in different ways. These kinds of systems constitute an extension of classical rule-based systems because they deal with fuzzy rules instead of classical logic rules.

Barelli and Bidini (2005) developed a diagnostic methodology for the optimization of energy efficiency and the maximization of the operational time based on artificial intelligence techniques such as artificial neural network (ANN) and fuzzy logic. The first part of the study deals with the identifying the principal modules and the corresponding variables necessary to evaluate the module "health state". Also the consequent upgrade of the monitoring system is described. Moreover it describes the structure proposed for the diagnostic procedure, consisting of a procedure for measurement validation and a fuzzy logic-based inference system. The first reveals the presence of abnormal conditions and localizes their source distinguishing between system failure and instrumentation malfunctions. The second provides an evaluation of module health state and the classification of the failures which have possibly occurred.

Wang and Dai (2004) propose a fuzzy constraint satisfaction approach to help buyers find fully satisfactory or replacement products in electronic shopping. For the buyer who can give precise product requirements, the proposed approach can generate product-ranking lists based on the satisfaction degrees of each product to the given requirements. For the buyer who may not input accurate requirements, a similarity analysis approach is proposed to assess buyer requirements automatically during his browsing process. The proposed approach could help buyers find the preferred products on the top of the ranking list without further searching the remaining pages.

Fang et al. (2004) propose a fuzzy set based formulation of auctions. They define fuzzy sets to represent the seller and buyers' valuations, bid possibilities and win possibilities. Analyzing the properties of these fuzzy sets, they study fuzzy versions of discriminating and nondiscriminating bidding strategies for each bidder in a single-object auction by using Bellman and Zadeh's concept of confluence of fuzzy decisions instead of the game theoretic Nash equilibrium. The monotonically decreasing membership function replaces of the corresponding distribution density function in Milgrom and Weber's auction theory. In the second part of the paper, they develop a soft computing approach to maximize the seller's revenue in multiple-object auctions through the use of object sequencing. The proposed sequencing approach is based on "fuzzy rule quantification".

Fonseca and Knapp (2001) develop a fuzzy reasoning algorithm and implement via an expert system to evaluate and assess the likelihood of equipment failure mode precipitation and aggravation. The scheme is based upon the fuzzification of the effects of precipitating factors provoking the failure. It consists of a fuzzy mathematical formulation which linearly relates the presence of factors catalogued as critical, important or related to the incidence of machine failure modes. This fuzzy algorithm is created to enable the inference mechanism of a constructed knowledgebased system to screen industrial equipment failures according to their likelihood of occurrence.

3.6 Artificial Intelligence

In order to model compex and sophisticated models artificial intelligence has the ability of combining several tools from several sources thereby, reducing the uncertainty. The most common known artificial intelligence techniques are ANN, fuzzy ANN, genetic algorithm and neuro-fuzzy approaches. Some of new studies on artificial intelligence are summarized in the following.

Palmero et al. (2005) introduce a system for fault detection and classification in AC motors based on soft computing. The kernel of the system is a neuro-fuzzy system, FasArt (Fuzzy Adaptive System ART-based), that permits the detection of a fault if it is in progress and its classification, with very low detection and diagnosis times that allow decisions to be made, avoiding definitive damage or failure when possible. The system is tested on an AC motor in which 15 nondestructive fault types were generated, achieving a high level of detection and classification. The knowledge stored in the neuro-fuzzy system is extracted by a fuzzy rule set with an acceptable degree of interpretability and without incoherency amongst the extracted rules.

Chang et al. (2005) present a fuzzy extension of the economic lot-size scheduling problem (ELSP) for fuzzy demands, since perturbations often occur in products demand in the real world. The ELSP is formulated via the extended basic period approach and power-of-two policy with the demands in triangular membership function. The resulting problem thus consists of a fuzzy total cost function, fuzzy feasibility constraints and a fuzzy continuous variable (the basic period) in addition to a set of crisp binary variables corresponding to the cycle times and starts of the products' schedule. Therefore, membership functions for the fuzzy total cost function and constraints can be figured out, from which the optimal fuzzy basic period and cycle times can be undetermined in addition to the compromised crisp values in fuzzy sense. Also, a genetic algorithm governed by the fuzzy total cost function and fuzzy feasibility constraints is designed and assists the ELSP in search for the optimal or near-optimal solution of the binary variables. This formulation is tested and illustrated on several ELSPs with varying levels of machine utilization and products demand perturbations. The results obtained are also analyzed with the lower bound generated by the independent solution approach to the ELSPs.

Gholamian et al. (2005) study an application of hybrid systematic design in multiobjective market problems. The target problem is suggested as unstructured real world problem such that the objectives cannot be expressed mathematically and only a set of historical data is utilized. Obviously, traditional methods and even meta-heuristic methods are broken in such cases. Instead, a systematic design using the hybrid of intelligent systems, particularly fuzzy rule base and neural networks can guide the decision maker towards noninferior solutions. The system does not stay in search phase. It also supports the decision maker in selection phase (after the search) to analyze various noninferior points and select the best ones based on the desired goal levels.

Kuo et al. (2005) propose a hybrid Case-Based Reasoning system with the integration of fuzzy sets theory and Ant System-based Clustering Algorithm (ASCA) in order to enhance the accuracy and speed in case matching. The cases in the case base are fuzzified in advance, and then grouped into several clusters by their own similarity with fuzzified ASCA. When a new case occurs, the system will find the closest group for the new case. Then the new case is matched using the fuzzy matching technique only by cases in the closest group. Through these two steps, if the number of cases is very large for the case base, the searching time will be dramatically saved. In the practical application, there is a diagnostic system for vehicle maintaining and repairing, and the results show a dramatic increase in searching efficiency.

Chen et al. (2004) develop an effective learning of neural network by using random fuzzy back-propagation learning algorithm. Based on this new learning algorithm, neural network not only has an accurate learning capability, but also can increase the probability of escaping from the local minimum while neural network is training. For demonstrating the new algorithm they develop has its outperformance, the classifications of the non-convex in two dimensions problem are simulated. For comparison, the same simulations by using conventional back-propagation learning algorithm with constant pairs of learning rate (D = 0.1-0.9) and momentum (x_i=0.1-0.9) and stochastic back propagation learning are also performed.

3.7 Engineering Economics

Engineering economics is the process involving mathematical analysis for selecting (or making a choice) from a set of alternatives based on clearly defined economic criteria. Many subtitles of engineering economics have been examined under fuzziness such as fuzzy capital budgeting techniques, fuzzy replacement analysis, etc.

Tolga et al. (2005) aim at creating an operating system selection framework for decision makers. Since decision makers have to consider both economic and non-economic aspects of technology selection, both factors have been considered in the developed framework. The economic part of the decision process is developed by Fuzzy Replacement Analysis. Noneconomic factors and financial figures are combined using a fuzzy analytic hierarchy process (Fuzzy AHP) approach. Since there exists incomplete and vague information of future cash flows and the crisp AHP cannot reflect the human thinking style in capturing the expert's knowledge, the fuzzy sets theory is applied to both AHP and replacement analysis, which compares two operating systems with and without license.

Chang (2005) presents a fuzzy methodology for replacement of equipment. Issues such as fuzzy modeling of degradation parameters and determining fuzzy strategic replacement and economic lives are extensively discussed. For the strategic purpose, addible market and cost effects from the replacements against the counterpart, i.e., the existing equipment cost and market obsolescence, are modeled fuzzily and interactively, in addition to the equipment deterioration. Both the standard fuzzy arithmetic and here re-termed requisite-constraint vertex fuzzy arithmetic (and the vertex method) are applied and investigated.

Kahraman et al. (2002) develop capital budgeting techniques using discounted fuzzy versus probabilistic cash flows the formulas for the analyses of fuzzy present value, fuzzy equivalent uniform annual value, fuzzy future value, fuzzy benefit-cost ratio, and fuzzy payback period are developed and given some numeric examples. Then the examined cash flows are expanded to geometric and trigonometric cash flows and using these cash flows fuzzy present value, fuzzy future value, and fuzzy annual value formulas are developed for both discrete compounding and continuous compounding.

Kuchta (2000) propose fuzzy equivalents of all the classical capital budgeting (investment choice) methods. These equivalents can be used to evaluate and compare projects in which the cash flows, duration time and required rate of return (cost of capital) are given imprecisely, in the form of a fuzzy number.

The application of discounted cash flow techniques for justifying manufacturing technologies is studied in many papers. State-price net present value and stochastic net present value are two examples of these applications. These applications are based on the data under certainty or risk. When we have vague data such as interest rate and cash flow to apply discounted cash flow techniques, the fuzzy set theory can be used to handle this vagueness. The fuzzy set theory has the capability of representing vague data and allows mathematical operators and programming to apply to the fuzzy domain. The theory is primarily concerned with quantifying the vagueness in human thoughts and perceptions. Assuming that they have vague data, Kahraman et al. (2000) use the fuzzy benefit-cost (B/C) ratio method to justify manufacturing technologies. After calculating the B/C ratio based on fuzzy equivalent uniform annual value, they compare two assembly manufacturing systems having different life cycles.

3.8 Ergonomics

Ergonomics is the science of adapting products and processes to human characteristics and capabilities in order to improve people's well-being and optimize productivity. The main topics of ergonomics in IE include anthropometry, human machine interaction and physical and mental health of workers. These topics are also studied by using the fuzzy logic techniques.

Kaya et al. (2004) take eighteen anthropometric measurements in standing and sitting positions, from 387 subjects between 15 and 17 years old. ANFIS is used to estimate anthropometric measurements as an alternative to stepwise regression analysis. Six outputs (shoulder width, hip width, knee height, buttock-popliteal height, popliteal height, and height) are selected for estimation purpose. The results show that the number of inputs required estimating outputs varied with sex difference. ANFIS perform better than stepwise regression method for both sex groups. Relevance to industry, more recently, as industry and marketing reach around the globe, body size has become a matter of practical interest to designers and engineers. In ergonomics anthropometric data are widely used to specify the physical dimension of workspaces, equipment, furniture and clothing. This is especially true for school children, which spend most of their time sitting at their chairs and desks and ought to be able to adopt comfortable body postures.

Park and Han (2004) propose a fuzzy rule-based approach to building models relating product design variables to affective user satisfaction. Affective user satisfaction such as luxuriousness, balance, and attractiveness are modeled for office chair designs. Regression models are also built on the same data to compare model performance. The results show that fuzzy rule-based models are better than regression models in terms of prediction performance and the number of variables included in the model. Methods for interpreting the fuzzy rules are discussed for practical applications in designing office chairs.

Bell and Crumpton (1997) present the development and evaluation of a fuzzy linguistic model designated to predict the risk of carpal tunnel syndrome (CTS) in an occupational setting. CTS is of the largest problems facing ergonomists and the medical community because it is developing in epidemic proportions within the occupational environment. In addition, practitioners are interested in identifying accurate methods for evaluating the risk of CTS in an occupational setting. It is hypothesized that many factors impact an individual's likelihood of developing CTS and the eventual development of CTS. This disparity in the occurrence of CTS for workers with similar backgrounds and work activities has confused researchers and has been a stumbling block in the development of a model for widespread use in evaluating the development of CTS. Thus this research is an attempt to develop a method that can be used to predict the likelihood of CTS risk in a variety of environments.

Work generally affects workers in terms of both physical and mental health. Workers must adapt their life pattern to match shift-work styles which can result in family problems, increased fatigue level, lower work efficiency, higher accident rate, illnesses, and lower productivity. Srithongchai and Intaranont (1996) obtain an entry permission to study the impact of shift work on fatigue level of workers in a sanitary-ware factory. The objectives are: 1) to evaluate fatigue levels of workers who work in the morning shift and night shift, and 2) to prioritize contributing factors affecting fatigue levels using a model of fuzzy set theory. Twelve male workers participate in the study. Four subjects are recruited from each of 3 departments, i.e., glazing, baking and quality inspection. The measurement is conducted before and after the shift for both shifts. Variables include heart rate monitoring throughout the work shift, critical flicker fusion frequency, reaction time response, hand-grip strength, and wet-bulb globe temperature. Results are analyzed using a computerized statistical package. It is concluded that mental fatigue from working in the morning shift is significantly higher than the one for working in the night shift. The same indication is also true in the case of physical fatigue, though it is not statistically significant. From the fuzzy set analysis, it is confirmed that working in the morning shift result in a higher fatigue level than working in the night shift and the temperature of the work environment is the most important factor contributing to the higher fatigue level.

Drastic changes in the nature of industrial work force, from manufacturing related industries to service and information industries, and increased automation have led to abundance of jobs for which there are no established performance standards. While work performance standards exist for physical tasks that are routinely performed in manufacturing industries, limits on acceptable levels of work are yet to be developed for mental tasks which are fast becoming common. Mital and Karwowski (1986) describe an approach to quantify qualitative human responses and describes a procedure to integrate them with other quantitative responses resulting from physical, mental, behavioral, or psycho-social factors. The theory of fuzzy sets and systems is utilized in developing the modeling concept which can be used in developing work-performance standards or optimizing overall man-machine systems effectiveness.

3.9 Production Planning and Control

Production systems form the base for building and improving the economic strength and vitality of a country. In order to manage production systems, production planning and control (PPC) techniques in various dimensions such as demand forecasting, aggregate planning, inventory management, material planning and operations scheduling, can be employed. In the literature very high number of papers on fuzzy PPC has been published. A summary of the direction of research on fuzzy production planning and control is found in the following.

Kasperski (2005) proposes a possibilistic approach to sequencing. For each parameter, whose value is not precisely known, a possibility distribution is given. The objective is to calculate a sequence of jobs, for which the possibility (necessity) of delays of jobs is minimal.

Chanas and Kasperski (2004) consider the single machine scheduling problem with parameters given in the form of fuzzy numbers is considered. It is assumed that the optimal schedule in such a problem cannot be determined precisely. They introduce the concepts of possible and necessary optimality of a given schedule. The degrees of possible and necessary optimality measure the possibility and the necessity of the event that a given schedule will be optimal. It is shown how to calculate the degrees of possible and necessary optimality of a given schedule in one of the special cases of the single machine scheduling problems.

Operation of complex shops needs specific, sophisticated procedures in order to guarantee competitive plant performance. Adenso-Díaz et al. (2004) present a hierarchy of models for roll shop departments in the steel industry, focusing on the calculation of the priority of the rolls to produce. A fuzzy-based model is developed and implemented in a real environment, allowing the simulation of expert behavior, considering the characteristics of an environment with imprecise information.

Majority of the products can be assembled in several ways that means the same final product can be realized by different sequences of assembly operations. Different degree of difficulty is associated with each sequence of assembly operation and such difficulties are caused by the different mechanical constraints forced by the different sequences of operations. In the past, few notable attempts have been made to represent and enumerate the degree of difficulty associated with an assembly sequence (in the form of triangular fuzzy number) by using the concept of assembly graph. However, such representation schemes do not possess the capabilities to model the user's reasoning and preferences. Ben-Arieh et al. (2004) present an intelligent Petri net model that combines the abilities of modeling, planning and performance evaluation for assembly operation. This modeling tool can represent the issues concerning degree of difficulty associated with assembly sequences. The proposed mechanism is enhanced expert high-level colored fuzzy Petri net that is a hybrid of knowledge-based system and colored Petri net.

Most of the studies on capacitated lot size problem are based on the assumption that the capacity is known exactly. However, in most practical applications, this is seldom the case. Fuzzy number theory is ideally suited to represent this vague and uncertain future capacity. Pai (2003) applies the fuzzy sets theory to solve this capacitated lot size problem.

Samanta and Al-Araimi (2001) propose a model based on fuzzy logic for inventory control. The periodic review model of inventory control with variable order quantity is considered. The model takes into account the dynamics of production-inventory system in a control theoretic approach. The control module combines fuzzy logic with proportional-integralderivative (PID) control algorithm. It simulates the decision support system to maintain the inventory of the finished product at the desired level in spite of variations in demand.

Singh and Mohanty (1991) characterize the manufacturing process plan selection problem as a machine routing problem. The routing problem is formulated as a multiple objective network model. Each objective is defined by a fuzzy membership function as a means of capturing the imprecision that exists when defining objectives. A dynamic programming solution procedure identifies the network path representing the best process plan. A dual-objective example is demonstrated for a component part requiring three machining operations. Two of the machining operations can be performed at alternative machining centers, resulting in a network model consisting of six nodes and eight branches. Cost and processing time per component are each represented by triangular fuzzy numbers. Zhang and Huang (1994) use fuzzy logic to model the process plan selection problem when objectives are imprecise and conflicting. Fuzzy membership functions are used to evaluate the contribution of competing process plans to shopfloor performance objectives. The optimal process plan for each part is determined by the solution of a fuzzy integer programming model. A consolidation procedure, which uses a dissimilarity criterion, then selects the process plan that best utilizes manufacturing resources. The algorithm is demonstrated for a problem consisting of three parts and eight process plans. The algorithm was also tested against non-fuzzy algorithms found in the literature. In some circumstances, more reasonable solutions where achieved as a result of the algorithm's ability to deal with the fuzziness inherent in manufacturing process planning. Inuiguchi *et al.* (1994) compare possibilistic, flexible and goal programming approaches to solving a production planning problem. Unlike conventional methods, possibilistic programming allows ambiguous data and objectives to be included in the problem formulation. A production planning problem consisting of two manufacturing processes, two products and four structural constraints is considered. The problem is solved using possibilistic programming, flexible programming and goal programming. A comparison of the three solutions suggests that the possibilistic solution best reflects the decision maker's input, thereby emphasizing the importance of modeling ambiguity in production planning.

Lee et al. (1991) extend their 1990 treatment of the MRP lot-sizing to include fuzzy modifications to the Silver-Meal, Wagner-Whitin, and partperiod balancing algorithms. The authors argue that when demands of the master schedule are truly fuzzy, demand should be modeled using membership functions. The performance of the three fuzzy lot-sizing algorithms is compared based on nine sample problems. Fuzzy set theory has been applied to problems in inventory management and production and process plan selection. The appeal of using fuzzy set theory in these IE problems echoes that of aggregate planning. Inventory management requires demand forecasts as ill as parameters for inventory related costs such as carrying, replenishment, shortages and backorders. Precise estimates of each of these model attributes are often difficult. Similarly, in production and process plan selection problems imprecision exists in specifying demand forecasts, inventory and processing cost parameters, processing times, and routing preferences. Potential ambiguity is further increased when the problem is formulated with multiple objectives. The research studies reviewed in this section demonstrate the usefulness of fuzzy set theory in modeling and solving inventory, and production and process plan selection problems when data and objectives are subject to ambiguity.

3.10 Aggregate Planning

Aggregate planning, sometimes called intermediate-range planning, involves production planning activities for six months to two year with monthly or quarterly updates. Changes in the workforce, additional machines, subcontracting, and overtime are typical decisions in aggregate planning.

Wang and Liang (2004) develop a fuzzy multi-objective linear programming (FMOLP) model for solving the multi-product aggregate production planning (APP) decision problem in a fuzzy environment. The proposed model attempts to minimize total production costs, carrying and backordering costs and rates of changes in labor levels considering inventory level, labor levels, capacity, warehouse space and the time value of money.

Wang and Fang (2001) present a novel fuzzy linear programming method for solving the aggregate production planning problem with multiple objectives where the product price, unit cost to subcontract, work force level, production capacity and market demands are fuzzy in nature. An interactive solution procedure is developed to provide a compromise solution.

Tang et al. (2000) focus on a novel approach to modeling multi-product APP problems with fuzzy demands and fuzzy capacities, considering that the demand requirements are fuzzy demand in each period during the planning horizon, The objective of the problem considered is to minimize the total costs of quadratic production costs and linear inventory holding costs. By means of formulation of fuzzy demand, fuzzy addition and fuzzy equation, the production inventory balance equation in single stage and dynamic balance equation are formulated as soft equations in terms of a degree of truth, and interpreted as the levels of satisfaction with production and inventory plan in meeting market demands. As a result, the multiproduct aggregate production planning problem with fuzzy demands and fuzzy capacities can be modeled into a fuzzy quadratic programming with fuzzy objective and fuzzy constraints.

Guiffrida and Nagi (1998) made a literature review on aggregate planning. Rinks (1981) cites a gap between aggregate planning theory and practice. Managers prefer to use their own heuristic decision rules over mathematical aggregate planning models. Using fuzzy conditional “ifthen” statements, Rinks develops algorithms for fuzzy aggregate planning. The strengths of the fuzzy aggregate planning model over traditional mathematical aggregate planning models include its ability to capture the approximate reasoning capabilities of managers, and the ease of formulation and implementation. The robustness of the fuzzy aggregate planning model under varying cost structures is examined in Rinks (1982a). A detailed set of forty production rate and work force rules is found in Rinks (1982b). Turksen (1988a, 1988b) advocates using interval-valued membership functions over the point-valued membership functions found in Rinks (1981, 1982a, 1982b) when defining linguistic production rules for aggregate planning. Ward *et al.* (1992) develop a C language program based on Rinks’ fuzzy aggregate planning framework. Gen *et al.* (1992) present a fuzzy multiple objective aggregate planning model. The model is formulated as a fuzzy multiple objective programming model with objective function coefficients, technological coefficients, and resource right-hand side values, represented by triangular fuzzy numbers. A transformation procedure is presented to transform the fuzzy multiple objective aggregate planning model into a crisp model. Aggregate planning involves the simultaneous determination of a company’s production, inventory, and workforce levels over a finite planning horizon such that total relevant cost is minimized.

Many aspects of the aggregate planning problem and the solution procedures employed to solve aggregate planning problems lend themselves to the fuzzy set theory approach. Fuzzy aggregate planning allows the vagueness that exists in the determining forecasted demand and the parameters associated with carrying charges, backorder costs, and lost sales to be included in the problem formulation. Fuzzy linguistic “if-then” statements may be incorporated into the aggregate planning decision rules as means for introducing the judgment and past experience of the decision maker into the problem. In this fashion, fuzzy set theory increases the model realism and enhances the implementation of aggregate planning models in industry. The usefulness of fuzzy set theory also extends to multiple objective aggregate planning models where additional imprecision due to conflicting goals may enter into the problem.

3.11 Quality Management

Research on fuzzy quality management is broken down into three areas, acceptance sampling, statistical process control, and general quality management topics. Quality management is a set of actions of the general management function which determines the quality policy, aims and responsibilities and realizes them within the framework of quality by planning, control, ensuring and improving the quality. An overview of research on fuzzy quality management is found in the following:

3.11.1 Acceptance Sampling

Ohta and Ichihashi (1988) present a fuzzy design methodology for single stage, two-point attribute sampling plans. An algorithm is presented and example sampling plans are generated when producer’s and consumer’s risk are defined by triangular fuzzy numbers. The authors do not address how to derive the membership functions for consumer’s and producer’s risk. Chakraborty (1988, 1994a) examines the problem of determining the sample size and critical value of a single sample attribute sampling plan when imprecision exists in the declaration of producer’s and consumer’s risk. In the 1988 paper, a fuzzy goal programming model and solution procedure are described. Several numerical examples are provided and the sensitivity of the strength of the resulting sampling plans is evaluated. The 1994a paper details how possibility theory and triangular fuzzy numbers are used in the single sample plan design problem.

Problems in the design of sampling inspection plans have been studied for a long time as an important subject in quality control. Especially, in the case of a single sampling attribute plan, one would like the producer's risk to equal D and the consumer's risk to equal E exactly, but this is usually not possible since the sample size *n* and the acceptance number *c* must be integers. Kanagawa and Ohta (1990) present a new design procedure for the single sampling attribute plan based on fuzzy sets theory by means of formulating this problem as a fuzzy mathematical programming one. Chakraborty (1992, 1994b) addresses the problem of designing single stage, Dodge-Romig lot tolerance percent defective (LTPD) sampling plans when the lot tolerance percent defective, consumer’s risk and incoming quality level are modeled using triangular fuzzy numbers. In the Dodge-Romig scheme, the design of an optimal LTPD sample plan involves solution to a nonlinear integer programming problem. The objective is to minimize average total inspection subject to a constraint based on the lot tolerance percent defective and the level of consumer’s risk. When fuzzy parameters are introduced, the procedure becomes a possibilistic (fuzzy) programming problem. A solution algorithm employing alpha-cuts is used to design a compromise LTPD plan, and a sensitivity analysis is conducted on the fuzzy parameters used.

3.11.2 Statistical Process Control

Cheng (2005) presents the construction of fuzzy control charts for a process with fuzzy outcomes derived from the subjective quality ratings provided by a group of experts. The proposed fuzzy process control methodology comprises an off-line stage and an on-line stage. In the off-line stage, experts assign quality ratings to products based on a numerical scale. The individual numerical ratings are then aggregated to form collective opinions expressed in the form of fuzzy numbers. The collective knowledge applied by the experts when conducting the quality rating process is acquired through a process of fuzzy regression analysis performed by a neural network. In the on-line stage, the product dimensions are measured, and the fuzzy regression model is employed to automate the experts' judgments by mapping the measured dimensions to appropriate fuzzy quality ratings. The fuzzy quality ratings are then plotted on fuzzy control charts, whose construction and out-of-control conditions are developed using possibility theory. The developed control charts not only monitor the central tendency of the process, but also indicate its degree of fuzziness.

Gülbay et al.’s (2004) approach provides the ability of determining the tightness of the inspection by selecting a suitable D-level. The higher D, the tighter inspection. The article also presents a numerical example and interprets and compares other results with the approaches developed previously.

Wang and Chen (1995) present a fuzzy mathematical programming model and solution heuristic for the economic design of statistical control charts. The economic statistical design of an attribute np-chart is studied under the objective of minimizing the expected lost cost per hour of operation subject to satisfying constraints on the Type I and Type II errors. The authors argue that under the assumptions of the economic statistical model, the fuzzy set theory procedure presented improves the economic design of control charts by allowing more flexibility in the modeling of the imprecision that exist when satisfying Type I and Type II error constraints.

Wang and Raz (1990) illustrate two approaches for constructing variable control charts based on linguistic data. When product quality can be classified using terms such as ‘perfect’, ‘good’, ‘poor’, etc., membership functions can be used to quantify the linguistic quality descriptions. Representative (scalar) values for the fuzzy measures may be found using the fuzzy mode or the alpha-level fuzzy midrange or the fuzzy median or the fuzzy average. The representative values that result from any of these methods are then used to construct the control limits of the control chart. Raz and Wang (1990) present a continuation of previous work on the construction of control charts for linguistic data. Results based on simulated data suggest that, on the basis of sensitivity to process shifts, control charts for linguistic data outperform conventional percentage defective charts. The number of linguistic terms used to represent the observation is found to influence the sensitivity of the control chart. Kanagawa *et al*. (1993) develop control charts for linguistic variables based on probability density functions which exist behind the linguistic data in order to control process average and process variability. This approach differs from the procedure of Wang and Raz in that the control charts are targeted at directly controlling the underlying probability distributions of the linguistic data.

Bradshaw (1983) uses fuzzy set theory as a basis for interpreting the representation of a graded degree of product conformance with a quality standard. When the costs resulting from substandard quality are related to the extent of nonconformance, a compatibility function exists which describes the grade of nonconformance associated with any given value of that quality characteristic. This compatibility function can then be used to construct fuzzy economic control charts on an acceptance control chart. The author stresses that fuzzy economic control chart limits are advantageous over traditional acceptance charts in that fuzzy economic control charts provide information on the severity as well as the frequency of product nonconformance.

3.11.3 General Topics in Quality Management

In view of the compatibility between psychology and linguistic terms, Tsai and Lu (2006) generalize the standard Choquet integral, whose measurable evidence and fuzzy measures are real numbers. The proposed generalization can deal with fuzzy-number types of measurable evidence and fuzzy measures. In comparison with the previous research on service quality and fuzzy sets, this study integrates the three-column format SERVQUAL into the generalized Choquet integral. A numerical example evaluating the overall service quality of e-stores is provided to illustrate how the generalized Choquet integral evaluates service quality in the three-column format SERVQUAL. This integral is also compared with the possibility/necessity model when measuring service quality.

Chien and Tsai (2000) propose a new method of measuring perceived service quality based on triangular fuzzy numbers. It avoids using difference scores (perceptions minus expectations) which are applied by many marketing researchers but are criticized by some other researchers. From consumers' standpoints, they replace perceptions by satisfaction degree as well as expectations by importance degree. To evaluate the discrepancy between consumers' satisfaction degree and importance degree, they induce general solutions to compute the intersection area between two triangular fuzzy numbers, then the weak and/or strong attributes of retail stores are clarified.

Kim et al. (2000) present an integrated formulation and solution approach to Quality Function Deployment (QFD). Various models are developed by defining the major model components (namely, system parameters, objectives, and constraints) in a crisp or fuzzy way using multiattribute value theory combined with fuzzy regression and fuzzy optimization theory. The proposed approach would allow a design team to reconcile tradeoffs among the various performance characteristics representing customer satisfaction as well as the inherent fuzziness in the system. In addition, the modeling approach presented makes it possible to assess separately the effects of possibility and flexibility inherent or permitted in the design process on the overall design. Knowledge of the impact of the possibility and flexibility on customer satisfaction can also serve as a guideline for acquiring additional information to reduce fuzziness in the system parameters as well as determine how much flexibility is warranted or possible to improve a design.

Khoo and Ho (1996) present a framework for a fuzzy quality function deployment (FQFD) system in which the 'voice of the customer' can be expressed as both linguistic and crisp variables. The FQFD system is used to facilitate the documentation process and consists of four modules (planning, deployment, quality control, and operation) and five supporting databases linked via a coordinating control mechanism. The FQFD system is demonstrated for determining the basic design requirements of a flexible manufacturing system. Glushkovsky and Florescu (1996) describe how fuzzy set theory can be applied to quality improvement tools when linguistic data is available. The authors identify three general steps for formalizing linguistic quality characteristics: (i) universal set choosing; (ii) definition and adequate formalization of terms; and (iii) relevant linguistic description of the observation. Examples of the application of fuzzy set theory using linguistic characteristics to Pareto analysis, cause-and-effect diagrams, design of experiments, statistical control charts, and process capability studies are demonstrated. Gutierrez and Carmona (1995) note that decisions regarding quality are inherently ambiguous and must be resolved based on multiple criteria. Hence, fuzzy multicriteria decision theory provides a suitable framework for modeling quality decisions. The authors demonstrate the fuzzy multiple criteria framework in an automobile manufacturing example consisting of five decision alternatives (purchasing new machinery, workforce training, preventative maintenance, supplier quality, and inspection) and four evaluation criteria (reduction of total cost, flexibility, leadtime, and cost of quality). Yongting (1996) identifies that failure to deal with quality as a fuzzy concept is a fundamental shortcoming of traditional quality management. Ambiguity in customers' understanding of standards, the need for multicriteria appraisal, and the psychological aspects of quality in the mind of the customer, support the modeling of quality using fuzzy set theory. A procedure for fuzzy process capability analysis is defined and is illustrated using an example. The application of fuzzy set theory in acceptance sampling, statistical process control and quality topics such as quality improvement and QFD has been reviewed in this section. Each of these areas requires a measure of quality. Quality, by its very nature, is inherently subjective and may lead to a multiplicity of meanings since it is highly dependent on human cognition. Thus, it may be appropriate to consider quality in terms of grades of conformance as opposed to absolute conformance or nonconformance. Fuzzy set theory supports subjective natural language descriptors of quality and provides a methodology for allowing them to enter into the modeling process. This capability may prove to be extremely beneficial in the further development of quality function deployment, process improvement tools and statistical process control.

3.12 Project Scheduling

Project scheduling is the task of planning timetables and the establishment of dates during which resources such as equipment and personnel, will perform the activities required to complete the project. In the following some recent works on fuzzy project scheduling are given.

Kim et al.'s (2003) develop a hybrid genetic algorithm (hGA) with fuzzy logic controller (FLC) to solve the resource-constrained project scheduling problem (rcPSP) which is a well-known NP-hard problem. Their new approach is based on the design of genetic operators with FLC and the initialization with the serial method, which has been shown superior for large-scale rcPSP problems. For solving these rcPSP problems, they firstly demonstrate that their hGA with FLC (flc-hGA) yields better results than several heuristic procedures presented in the literature. Then they evaluate several genetic operators, which include compounded partially mapped crossover, position-based crossover, swap mutation, and local search-based mutation in order to construct the flc-hGA which has the better optimal makespan and several alternative schedules with optimal makespan.

Wang (2002) develop a fuzzy scheduling methodology to deal with this problem. Possibility theory is used to model the uncertain and flexible temporal information. The concept of schedule risk is proposed to evaluate the schedule performance. A fuzzy beam search algorithm is developed to determine a schedule with the minimum schedule risk and the start time of each activity is selected to maximize the minimum satisfaction degrees of all temporal constraints. In addition, the properties of schedule risk are also discussed. They show that the proposed methodology can assist project managers in selecting a schedule with the least possibility of being late in an uncertain scheduling environment.

Hapke and Slowinski (1996) present a generalization of the known priority heuristic method for solving resource-constrained project scheduling problems (RCPS) with uncertain time parameters. The generalization consists of handling fuzzy time parameters instead of crisp ones. In order to create priority lists, a fuzzy ordering procedure has been proposed. The serial and parallel scheduling procedures which usually operate on these lists have also been extended to handle fuzzy time parameters.

In the following Guiffrida and Nagi's (1998) literature survey on project scheduling is given. Notice that the majority of the research on this topic has been devoted to fuzzy PERT. Prade (1979) applies fuzzy set theory to the development of an academic quarter schedule at a French school. When data are not precisely known, fuzzy set theory is shown to be relevant to the exact nature of the problem rather than probabilistic PERT or CPM. The aim of this work is to show how and when it is possible to use fuzzy concepts in a real world scheduling problem. An overview of a fuzzy modification to the classic Ford solution algorithm is presented along with a 17 node network representation for the academic scheduling problem. Calculations are demonstrated for a small portion of the overall scheduling problem. Chanas and Kamburowski (1981) argue the need for an improved version of PERT due to three circumstances: (i) the subjectivities of activity time estimates; (ii) the lack of repeatability in activity duration times; and (iii) calculation difficulties associated with using probabilistic methods. A fuzzy version of PERT (FPERT) is presented in which activity times are represented by triangular fuzzy numbers. Kaufmann and Gupta (1988) devote a chapter of their book to the critical path method in which activity times are represented by triangular fuzzy numbers. A six step procedure is summarized for developing activity estimates, determining activity float times, and identifying the critical path. A similar tutorial on fuzzy PERT involving trapezoidal fuzzy numbers may be found in Dubois and Prade (1985). McCahon and Lee (1988) note that PERT is best suited for project network applications when past experience exists to allow the adoption of the beta distribution for activity duration times and when the network contains approximately 30 or more activities. When activity times are vague, the project network should be modeled with fuzzy components. A detailed example demonstrates modeling and solving an eight activity project network when activity durations are represented as triangular fuzzy numbers. Lootsma (1989) identifies that human judgment plays a dominant role in PERT due to the estimation of activity durations and the requirement that the resulting plan be tight. This aspect of PERT exposes the conflict between normative and descriptive modeling approaches. Lootsma argues that vagueness is not properly modeled by probability theory, and rejects the use of stochastic models in PERT planning when activity durations are estimated by human experts. Despite some limitations inherent in the theory of fuzzy sets, fuzzy PERT, in many respects, is closer to reality and more workable than stochastic PERT. Buckley (1989) provides detailed definitions of the possibility distributions and solution algorithm required for using fuzzy PERT. A ten activity project network example in which activity durations are described by triangular fuzzy numbers, is used to demonstrate the development of the possibility distribution for the project duration. Possibility distributions for float, earliest start, and latest start times are defined, but not determined, due to their complexity. DePorter and Ellis (1990) present a project crashing model using fuzzy linear programming. Minimizing project completion time and project cost are highly sought yet conflicting project objectives. Linear programming allows the optimization of one objective (cost or time). Goal programming allows consideration of both time and cost objectives in the optimization scheme. When environmental factors present additional vagueness, fuzzy linear programming should be used. Linear programming, goal programming and fuzzy linear programming are applied to a ten activity project network. Project crashing costs and project durations are determined under each solution technique. McCahon (1993) compares the performance of fuzzy project network analysis (FPNA) and PERT. Four basic network configurations were used. The size of the networks ranged from four to eight activities. Based on these networks, a total of thirty-two path completion times were calculated using FPNA and PERT. The performance of FPNA and PERT was compared using: the expected project completion time, the identification of critical activities, the amount of activity slack, and the possibility of meeting a specified project completion time. The results ofthis study conclude that PERT estimates FPNA adequately. When estimating expected project completion time however, a generalization concerning compared performance with respect to the set of critical activities, slack times and possibility of project completion times cannot be made. When activity times are poorly defined, the performance of FPNA outweighs its cumbersomeness and should be used instead of PERT. Nasution (1994) argues that for a given alpha-cut level of the slack, the availability of the fuzzy slack in critical path models provides sufficient information to determine the critical path. A fuzzy procedure utilizing interactive fuzzy subtraction is used to compute the latest allowable time and slack for activities. The procedure is demonstrated for a ten event network where activity times are represented by trapezoidal fuzzy numbers. Hapke (1994) present a fuzzy project scheduling (FPS) decision support system. The FPS system is used to allocate resources among dependent activities in a software project scheduling environment. The FPS system uses L-R type flat fuzzy numbers to model uncertain activity durations. Expected project completion time and maximum lateness are identified as the project performance measures and a sample problem is demonstrated for a software engineering project involving 53 activities. The FPS system presented allows the estimation of Project completion times and the ability to analyze the risk associated with overstepping the required project completion time. Lorterapong (1994) introduces a resource-constrained project scheduling method that addresses three performance objectives: (i) expected project completion time; (ii) resource utilization; and (iii) resource interruption Fuzzy set theory is used to model the vagueness that is inherent with linguistic descriptions often used by people when describing activity durations. The analysis presented provides a framework for allocating resources in an uncertain project environment. Chang *et al.* (1995) combine the composite and comparison methods of analyzing fuzzy numbers into an efficient procedure for solving project scheduling problems. The comparison method first eliminates activities that are not on highly critical paths. The composite method then determines the path with the highest degree of criticality. The fuzzy Delphi method (see Kaufmann and Gupta (1988)) is used to determine the activity time estimates. The solution procedure is demonstrated in a 9 node, 14 activity project scheduling problem with activity times represented by triangular fuzzy numbers. Shipley *et al.* (1996) incorporate fuzzy logic, belief functions, extension principles and fuzzy probability distributions, and developed the fuzzy PERT algorithm, ‘Belief in Fuzzy Probabilities of Estimate Time’ (BIFPET). The algorithm is applied to a real world project consisting of eight activities involved in the selling and producing of a 30-second television commercial. Triangular fuzzy numbers are used to define activity durations. BIFPET is used to determine the project critical path and expected project completion time. The specification of activity duration times is crucial to both CPM and PERT project management applications. In CPM, historical data on the duration of activities in exact or very similar projects exists, and it is used to specify activity durations for similar future projects. In new projects where no historical data on activity durations exists, PERT is often used. Probabilisticbased PERT requires the specification of probability distribution (frequently the beta distribution) to represent activity durations. Estimates of the first two moments of the beta distribution provide the mean and variance of individual activity durations. Fuzzy set theory allows the human judgment that is required when estimating the behavior of activity durations to be incorporated into the modeling effort. The versatility of the fuzzy-theoretic approach is further championed in resource constrained and project crashing scenarios where additional uncertainty is introduced when estimating resource availability and cost parameters.

3.13 Facility Location and Layout

Facility location problems consist of choosing when and where to open facilities in order to minimize the associated cost of opening a facility and the cost of servicing customers. The design of the facility layout of a manufacturing system is critically important for its effective utilization: 20 to 50 percent of the total operating expenses in manufacturing are attributed to material handling and layout related costs. Use of effective methods for facilities planning can reduce these costs often by as much as 30 percent, and sometimes more. The problems of facility location and layout have been studied extensively in the industrial engineering literature.

3.13.1 Facility Location

Gen and Syarif (2005) deal with a production/distribution problem to determine an efficient integration of production, distribution and inventory system so that products are produced and distributed at the right quantities, to the right customers, and at the right time, in order to minimize system wide costs while satisfying all demand required. This problem can be viewed as an optimization model that integrates facility location decisions, distribution costs, and inventory management for multi-products and multi-time periods. To solve the problem, they propose a new technique called spanning tree-based genetic algorithm (hst-GA). In order to improve its efficiency, the proposed method is hybridized with the fuzzy logic controller (FLC) concept for auto-tuning the GA parameters.

Kahraman et al. (2003) solve facility location problems using different solution approaches of fuzzy multi-attribute group decision-making. Their paper includes four different fuzzy multi-attribute group decision-making approaches. The first one is a fuzzy model of group decision proposed by Blin. The second is the fuzzy synthetic evaluation. The third is Yager's weighted goals method and the last one is fuzzy analytic hierarchy process. Although four approaches have the same objective of selecting the best facility location alternative, they come from different theoretic backgrounds and relate differently to the discipline of multi-attribute group decisionmaking. These approaches are extended to select the best facility location alternative by taking into account quantitative and qualitative criteria. A short comparative analysis among the approaches and a numeric example to each approach are given.

Chu (2002) presents a fuzzy TOPSIS model under group decisions for solving the facility location selection problem, where the ratings of various alternative locations under different subjective attributes and the importance weights of all attributes are assessed in linguistic values represented by fuzzy numbers. The objective attributed are transformed into dimensionless indices to ensure compatibility with the linguistic ratings of the subjective attributes. Furthermore, the membership function of the aggregation of the ratings and weights for each alternative location versus each attribute can be developed by interval arithmetic and Į-cuts of fuzzy numbers. The ranking method of the mean of the integral values is applied to help derive the ideal and negative-ideal fuzzy solutions to complete the proposed fuzzy TOPSIS model.

Guiffrida and Nagi (1998) made a literature review on facility location. Narasimhan (1979) presents an application of fuzzy set theory to the problem of locating gas stations. Fuzzy ratings are used to describe the relative importance of eleven attributes for a set of three location alternatives. A Delphi-based procedure was applied, and the input of decision makers was used to constructmembership functions for three importance weights for judging attributes. Computations are summarized for the selection decision. The author concludes that the procedure presented is congruent to the way people make decisions. The procedure provides a structure for organizing information, and a systematic approach to the evaluation of imprecise and unreliable information. Darzentas (1987) formulates the facility location problem as a fuzzy set partitioning model using integer programming. This model is applicable when the potential facility points are not crisp and can best be described by fuzzy sets. Linear membership functions are employed in the objective function and constraints of the model. The model is illustrated with an example based on three location points and four covers. Mital et al. (1988) and Mital and Karwowski (1989) apply fuzzy set theory in quantifying eight subjective factors in a case study involving the location of a manufacturing plant. Linguistic descriptors are used to describe qualitative factors in the location decision, such as community attitude, quality of schools, climate, union attitude, nearness to market, police protection, fire protection, and closeness to port. Bhattacharya et al. (1992) present a fuzzy goal programming model for locating a single facility within a given convex region subject to the simultaneous consideration of three criteria: (i) maximizing the minimum distances from the facility to the demand points; (ii) minimizing the maximum distances from the facilities to the demand points; and (iii) minimizing the sum of all transportation costs. Rectilinear distances are used under the assumption that an urban scenario is under investigation. A numerical example consisting with three demand points is given to illustrate the solution procedure. Chung and Tcha (1992) address the location of public supply-demand distribution systems such as a water supply facility or a waist disposal facility. Typically, the location decision in these environments is made subject to the conflicting goals minimization of expenditures and the preference at each demand site to maximizing the amount supplied. A fuzzy mixed 0-1 mathematical programming model is formulated to study both uncapacitated and capacitated modeling scenarios. The objective function includes the cost of transportation and the fixed cost for satisfying demand at each site. Each cost is represented by a linear membership function. Computational results for twelve sample problems are demonstrated for a solution heuristic based on Erlenkotter's dual-based procedure for the uncapacitated facility location problem. Extension to the capacitated case is limited by issues of computational complexity and computational results are not presented. Bhattacharya et al. (1993) formulate a fuzzy goal programming model for locating a single facility within a given convex region subject to the simultaneous consideration of two criteria: (i) minimize the sum of all transportation costs; and (ii) minimize the maximum distances from the facilities to the demand points. Details and assumptions of the model are similar to Bhattacharya et al. (1992). A numerical example consisting of two facilities and three demand points is presented and solved using LINDO.

3.13.2 Facility Layout

Dweiri (1999) presents a distinct methodology to develop a crisp activity relationship charts based on fuzzy set theory and the pairwise comparison of Saaty's AHP, which ensures the consistency of the designer's assignments of importance of one factor over another to find the weight of each of the factors in every activity.

Deba and Bhattacharyyab (2005) present a distinct decision support system based on multifactor fuzzy inference system (FIS) for the development of facility layout with fixed pickup/drop-off points. The algorithm searches several candidate points with different orientation of incoming machine blocks in order to minimize flow cost, dead space and area required for the development of layout.

Ertay et al. (2006) presents a decision-making methodology based on data envelopment analysis (DEA), which uses both quantitative and qualitative criteria, for evaluating facility layout design. The criteria that are to be minimized are viewed as inputs.

Guiffrida and Nagi (1998) made a literature review on facility layout. Grobelny (1987a, 1987b) incorporates the use of 'linguistic patterns' in solving the facility layout problem. For example, if the flow of materials between departments is high, then the departments should be located close to each other. The linking between the departments and the distance between the departments represent linguistic (fuzzy) variables; the 'high' and 'close' qualifications represent values of the linguistic variables. The evaluation of a layout is measured as the grade of satisfaction as measured by the mean truth value, of each linguistic pattern by the final placement of departments. Evans *et al.* (1987) introduce a fuzzy set theory based construction heuristic for solving the block layout design problem. Qualitative layout design inputs of 'closeness' and 'importance' are modeled using linguistic variables. The solution algorithm selects the order of department placement which is manual. The algorithm is demonstrated by determining a layout for a six department metal fabrication shop. The authors identify the need for future research toward the development of a heuristic that address both the order and placement of departments, the selection of values for the linguistic variables, and the determination of membership functions. Raoot and Rakshit (1991) present a fuzzy layout construction algorithm to solve the facility layout problem. Linguistic variables are used in the heuristic to describe qualitative and quantitative factors that affect the layout decision. Linguistic variables capture information collected from experts for the following factors: flow relationships, control relationships, process and service relationships, organizational and personnel relationships, and environmental relationships. Distance is also modeled as a fuzzy variable and is used by the heuristic as the basis for placement of departments. Three test problems are used to compare the fuzzy heuristic with ALDEP and CORELAP. The authors note that the differences achieved by each of the three methods is a function of the different levels of reality that they use. Raoot and Rakshit (1993) formulate the problem of evaluating alternative facility layouts as a multiple criteria decision model (MCDM) employing fuzzy set theory. The formulation addresses the layout problem in which qualitative and quantitative factors are equally important. Linguistic variables are used to capture experts' opinions regarding the primary relationships between departments. Membership functions are selected based on consultation with layout experts. The multiple objectives and constraints of the formulation are expressed as linguistic patterns. The fuzzy MCDM layout algorithm is demonstrated for the layout of an eight department facility. Raoot and Rakshit (1994) present a fuzzy set theory-based heuristic for the multiple goal quadratic assignment problem The objective function in this formulation utilizes ‘the mean truth value’, which indicates the level of satisfaction of a layout arrangement to the requirements of the layout as dictated by a quantitative or qualitative goal. The basic inputs to the model are expert’s opinions on the qualitative and quantitative relationships between pairs of facilities. The qualitative and quantitative relationships are captured by linguistic variables, membership functions are chosen arbitrarily. Three linguistic patterns (one quantitative; two qualitative) are employed by the heuristic to locate facilities. The performance of the heuristic is tested against a set of test problems taken from the open literature. The results of the comparison indicate that the proposed fuzzy heuristic performs well in terms of the quality of the solution. Dweiri and Meier (1996) define a fuzzy decision making system consisting of four principal components: (i) fuzzification of input and output variables; (ii) the experts’ knowledge base; (iii) fuzzy decision making; and (iv) defuzzification of fuzzy output into crisp values. The analytical hierarchy process is used to weigh factors affecting closeness ratings between departments.

3.14 Forecasting

Forecasting methods are used to predict future events in order to make good decisions. Since the subject is about predicting and foreseeing, it is unavoidable to encounter uncertainty or vagueness. Therefore fuzzy set theory is intensively applied to forecasting issues. In the following, first Guiffrida and Nagi’s (1998) literature review is given and then some recent works are summarized.

The first application of forecasting using fuzzy set theory to our knowledge appeared in Economakos (1979). A simulation based model was used to forecast the demand for electrical power when load components at various times of the day were described in linguistic terms. Interest in fuzzy forecasting has grown considerably since this initial article. Research on fuzzy forecasting is divided into three categories: qualitative forecasting employing the Delphi method, time series analysis and regression analysis. A summary of fuzzy forecasting applications is presented in the following.

Murray *et al.* (1985) identify fuzzy set theory as an attractive methodology for modeling ambiguity in the Delphi method. Ambiguity results due to differences in the meanings and interpretations that experts may attach to words. A pilot Delphi study using graduate business students (divided into control and test groups), as experts, was conducted to estimate the percentage of students attaining an “excellent” grade point average. Over four rounds of the Delphi, the control group received the typical feedback consisting of a summary of their responses. The test group received similar feedback but was also given feedback on a group average membership function. Methodological issues inherent in the study prevent a statistical analysis of the performance of the control and test groups with respect to the Delphi task. However, the authors demonstrate an appropriate approach for modeling ambiguity in the Delphi method, and provide insights on methodological issues for future research. Kaufmann and Gupta (1988) present a detailed tutorial on the fuzzy Delphi method in forecasting and decision making. The authors outline a four-step procedure for performing fuzzy Delphi when estimates are represented by triangular fuzzy numbers. Fuzzy Delphi is demonstrated by example for a group of twelve experts engaged in forecasting the realization of a cognitive information processing computer.

Ishikawa *et al.* (1993) cite limitations in traditional and fuzzy Delphi methods and propose the New Fuzzy Delphi Method (NFDM). The NFDM has the following advantages: (i) fuzziness is inescapably incorporated in the Delphi findings; (ii) the number of rounds in the Delphi is reduced; (iii) the semantic structure of forecast items is refined; and (iv) the individual attributes of the expert are clarified. The NFDM consists of two methodologies: the Min-Max Delphi Method, and the Fuzzy Delphi Method via Fuzzy Integration. The Max-Min Delphi Method clarifies data of each forecaster by expertise, pursues the accuracy of the forecast from the standpoint of an interval representing possibility and impossibility, and identifies the cross point as the most attainable period. The Fuzzy Delphi Method via Fuzzy Integration employs the expertise of each expert as a fuzzy measure and identifies a point estimate as the most attainable period by the fuzzy integration of each membership function. The two methodologies are illustrated by way of a technological forecasting example using members of the Japan Society for Fuzzy Theory and Systems as experts.

Song and Chissom (1993a) provide a theoretic framework for fuzzy time series modeling. A fuzzy time series is applicable when a process is dynamic and has historical data that are fuzzy sets or linguistic values. Fuzzy relational equations are employed to develop fuzzy relations among observations occurring at different time periods. Two classes of fuzzy time series models are defined: time-variant and time-invariant. A seven-step procedure is outlined for conducting a forecast using the time-invariant fuzzy time series model. Song and Chissom (1993b) apply a first order, time-invariant time series model to forecast enrollments of the University of Alabama based on twenty years of historical data. The data was fuzzified and seven fuzzy sets were defined to describe “enrollments”. The corresponding linguistic values ranged from “not many” to “too many”. The seven-step procedure outlined in Song and Chissom (1993a) is used to fuzzify the data, develop the time series model, and calculate and interpret the output. The errors in the forecasted enrollments ranged from 0.1% to 11%, with an average error of 3.9%. The error resulting from the fuzzy time series model is claimed to be on par with error rates cited in the literature on enrollment forecasting.

Cummins and Derrig (1993) present a fuzzy-decision making procedure for selecting the best forecast subject to a set of vague or fuzzy criteria. Selecting the best forecast, as opposed to selecting the best forecasting model, most efficiently utilizes the historical information available to the forecaster. Fuzzy set theory is used to rank candidate forecasting methods in terms of their membership values in the fuzzy set of “good” forecasts. The membership values are then used to conclude the best forecast. The methodology is demonstrated in developing a forecast of a trend component for an insurance loss cost problem. A set of 72 benchmark forecasting methods are consolidated using fuzzy set theory. On the basis of three fuzzy objectives, a fuzzy set of “good” trend factors results. The authors conclude that fuzzy set theory provides an effective method for combining statistical and judgmental criteria in actuarial decision making. Song and Chissom (1994) apply a first order, time-variant forecasting model to the Song and Chissom(1993b) university enrollment data set. The time variant-model relaxes the invariant-model assumption that at any time t, the possible values of the fuzzy time series are the same. A 3-layer backpropagation neural network was found to be the most effective method for defuzzifying the forecast fuzzy set. The neural network defuzzification method yielded the smallest average forecasting error over a range of model structures. Sullivan and Woodall (1994) provide a detailed review of the time-invariant and time-variant time series models set forth by Song and Chissom (1993a, 1994). The authors present a Markov model for forecasting enrollments. The Markov approach utilizes linguistic labels and uses probability distributions, as opposed to membership functions, to reflect ambiguity. Using the enrollment data set of Song and Chissom, a Markov model is described and the parameter estimation procedure is compared with that of the fuzzy time series method. The Markov approach resulted in slightly more accurate forecasts than the fuzzy time series approach. The Markov and fuzzy time series models are compared to three conventional time-invariant time series models, a first-order autoregressive model, and two second-order auto- regressive models. The conventional models which use the actual crisp enrollment data, outperformed the fuzzy time series or Markov model and are most effective when predicting values outside the range of the historical data. Song *et al.* (1995) note that the fuzzy time series models defined in Song and Chissom (1993a), and those used to forecast university enrollments in Song and Chissom (1993b, 1994), require fuzzification of crisp historical data. The authors present a new model based on the premise that the historical data are fuzzy numbers. The new model is based on a theorem derived to relate the current value of a fuzzy time series with its past. The theorem expresses the value of a homogeneous fuzzy time series as a linguistic summation of previous values and the linguistic differences of different orders. A second form of the model relates the value of the fuzzy time series as the linguistic summation of various order linguistic backward differences. The authors note that the findings of this study are limited only to homogeneous fuzzy time series with fuzzy number observations. Chen (1996) presents a modified version Song and Chissom’s time-invariant fuzzy time series model. The modified model is claimed to be “obviously” more efficient than Song and Chissom's (1993b) model. The claim is based on the proposed model's use of simplified arithmetic operations rather than the max-min composition operators found in Song and Chissom's model. Forecasts based on the modified model and Song and Chissom's model are compared to the historical enrollment data. The forecasted results of the proposed model deviate slightly from Song and Chissom. The statistical significance of the comparison is not addressed.

Tanaka *et al.* (1982) introduce fuzzy linear regression as a means to model casual relationships in systems when ambiguity or human judgment inhibits a crisp measure of the dependent variable. Unlike conventional regression analysis, where deviations between observed and predicted values reflect measurement error, deviations in fuzzy regression reflect the vagueness of the system structure expressed by the fuzzy parameters of the regression model. The fuzzy parameters of the model are considered to be possibility distributions which correspond to the fuzziness of the system. The fuzzy parameters are determined by a linear programming procedure which minimizes the fuzzy deviations subject to constraints of the degree of membership fit. The fuzzy regression forecasting model is demonstrated by a multiple regression example in which the prices of prefabricated houses is determined based on quality of material, floor space, and number of rooms. Heshmaty and Kandel (1985) utilize fuzzy regression analysis to build forecasting models for predicting sales of computers and peripheral equipment. The independent variables are: user population expansion, microcomputer sales, minicomputer sales, and the price of microcomputers. The forecasts for the sales of computers and peripheral equipment are given as fuzzy sets with triangular membership functions. The decision-maker then selects the forecast figure from within the interval of the fuzzy set. Fuzzy set theory has been used to develop quantitative forecasting models such as time series analysis and regression analysis, and in qualitative models such as the Delphi method. In these applications, fuzzy set theory provides a language by which indefinite and imprecise demand factors can be captured. The structure of fuzzy forecasting models are often simpler yet more realistic than non-fuzzy models which tend to add layers of complexity when attempting to formulate an imprecise underlying demand structure. When demand is definable only in linguistic terms, fuzzy forecasting models must be used.

Chang (1997) presents a fuzzy forecasting technique for seasonality in the time-series data using the following procedure. First, with the fuzzy regression analysis the fuzzy trend of a time-series is analyzed. Then the fuzzy seasonality is defined by realizing the membership grades of the seasons to the fuzzy regression model. Both making fuzzy forecast and crisp forecast are investigated. Seasonal fuzziness and trends are analyzed.

Chen and Wang (1999) apply fuzzy concepts in forecasting product price and sales in the semiconductor industry which is often conceived as a highly dynamic environment. First, two fuzzy forecasting methods including fuzzy interpolation and fuzzy linear regression are developed and discussed. Forecasts generated by these methods are fuzzy-valued. Next, the subjective beliefs about whether the industry is booming or slumping and the speed at which this change in prosperity takes place during a given period are also considered. Two subjective functions are defined and used to adjust fuzzy forecasts. Practically, fuzzy forecasts are incorporated with fuzzy programming like fuzzy linear programming or fuzzy nonlinear programming for mid-term or long-term planning.

In modeling a fuzzy system with fuzzy linear functions, the vagueness of the fuzzy output data may be caused by both the indefiniteness of model parameters and the vagueness of the input data. This situation occurs as the input data are envisaged as facts or events of an observation which are uncontrollable or uninfluenced by the observer rather than as the controllable levels of factors in an experiment. Lee and Chen (2001) concentrate on such a situation and refer to it as a generalized fuzzy linear function. Using this generalized fuzzy linear function, a generalized fuzzy regression model is formulated. A nonlinear programming model is proposed to identify the fuzzy parameters and their vagueness for the generalized regression model. A manpower forecasting problem is used to demonstrate the use of the proposed model.

Kuo (2001) utilizes a fuzzy neural network with initial weights generated by genetic algorithm (GFNN) for the sake of learning fuzzy IF-THEN rules for promotion obtained from marketing experts. The result from GFNN is further integrated with an ANN forecast using the time series data and the promotion length from another ANN. Model evaluation results for a convenience store (CVS) company indicate that the proposed system can perform more accurately than the conventional statistical method and a single ANN.

Sales forecasting plays a very prominent role in business strategy. Numerous investigations addressing this problem have generally employed statistical methods, such as regression or autoregressive and moving average (ARMA). However, sales forecasting is very complicated owing to influence by internal and external environments. Recently, ANNs have also been applied in sales forecasting since their promising performances in the areas of control and pattern recognition. However, further improvement is still necessary since unique circumstances, e.g. promotion, cause a sudden change in the sales pattern. Kuo et al. (2002) utilize a proposed fuzzy neural network (FNN), which is able to eliminate the unimportant weights, for the sake of learning fuzzy IF-THEN rules obtained from the marketing experts with respect to promotion. The result from FNN is further integrated with the time series data through an ANN. Both the simulated and realworld problem results show that FNN with weight elimination can have lower training error compared with the regular FNN. Besides, real-world problem results also indicate that the proposed estimation system outperforms the conventional statistical method and single ANN in accuracy.

Reliable prediction of sales can improve the quality of business strategy. Chang and Wang (2005) integrate fuzzy logic and artificial neural network into the fuzzy back-propagation network (FBPN) for sales forecasting in printed circuit board industry. The fuzzy back propagation network is constructed to incorporate production-control expert judgments in enhancing the model's performance. Parameters chosen as inputs to the FBPN are no longer considered as of equal importance, but some sales managers and production control experts are requested to express their opinions about the importance of each input parameter in predicting the sales with linguistic terms, which can be converted into pre-specified fuzzy numbers.

4 Conclusions

This chapter has discussed an extensive literature review and survey of fuzzy set theory in industrial engineering. Throughout the course of this study, it has been observed that (1) fuzzy set theory has been applied to most traditional areas of industrial engineering, and (2) research on fuzzy set theory in industrial engineering has grown in recent years. Fuzzy research in quality management, forecasting, and job shop scheduling have experienced tremendous growth in recent years. The appropriateness and contribution of fuzzy set theory to problem solving in industrial engineering may be seen by paralleling its use in operations research. Zimmerman (1983) identifies that fuzzy set theory can be used in operations research as a language to model problems which contain fuzzy phenomena or relationships, as a tool to analyze such models in order to gain better insight into the problem and as an algorithmic tool to make solution procedures more stable or faster. We hope this chapter should give industrial engineering researchers new tools and ideas on how to approach industrial engineering problems using fuzzy set theory. It also provides a basis for fuzzy set researchers to expand on the toolset for industrial engineering problems.

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